

Perceptually Intelligent UI for Augmented Everyday Interaction

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Abstract

Extended Reality (XR) systems promise to seamlessly integrate digital information into physical environments, but this integration creates fundamental challenges as digital content can interfere with real-world activities and compete for attention. My research addresses these challenges by developing **perceptually intelligent user interfaces** that automatically adapt the representation to users' perceptual capabilities, situational constraints, and evolving contexts. Through three interconnected thrusts—(1) modeling multisensory perception and behavior, (2) developing context-aware UI representations, and (3) supporting persistent interaction continuity—I demonstrate computational approaches that predict sensory confusion and correspondence; generate fluid, recomposable interfaces; and maintain awareness of user history and intent across time and space. These approaches effectively improve seamless access to digital tools by optimizing for accurate and intuitive perception while minimizing clutter and overload. I conclude by discussing future directions toward perceptually intelligent UI, building on these foundations.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Interaction techniques**.

Keywords

computational modeling, extended reality, multisensory interfaces

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1 Introduction

Extended Reality (XR) devices promise to revolutionize how users interact with digital information by seamlessly embedding it into our physical surroundings—helping users navigate unfamiliar places, remember important details, and take action without breaking stride. Unlike screen-based computing, which requires users to shift attention away from their environment, XR can support users

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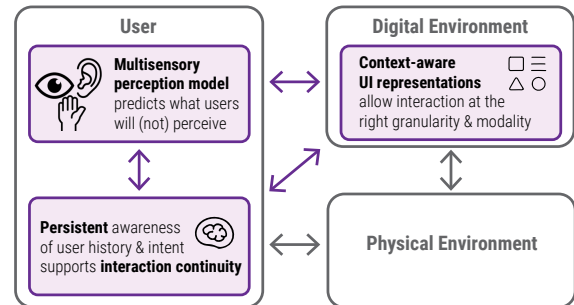


Figure 1: Three interconnected research thrusts toward perceptually intelligent UI.

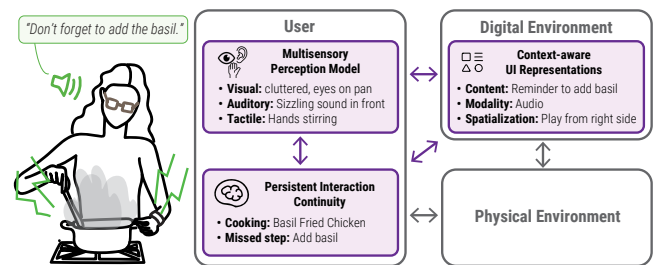


Figure 2: Example of a perceptually intelligent UI adapting to a cooking scenario. It infers the user's current perceptual state across visual, auditory, and tactile channels, recognizes a missed step ("add basil"), and selects the appropriate modality and spatial audio placement to deliver a just-in-time reminder—minimizing overload and ambiguity.

while they interact with the world. Yet, realizing this vision carries fundamental challenges. As XR systems inhabit the same physical space as users, digital content can interfere with real-world activities and compete for attention during critical moments.

This stems from three key limitations. First, XR systems lack awareness of users' perceptual states, resulting in missed cues or overload. Second, XR interfaces inherit rigid app structures from screen-based computing, which assume focused attention and stable contexts, creating friction when users are cognitively occupied or physically active. Third, interactions are treated as isolated episodes, forcing users to repeatedly re-establish context and intent, which disrupts the continuous flow of real-world activities.

To address these challenges, my research envisions **perceptually intelligent user interfaces** (Figure 1) that adapt to the user's perceptual, situational, and cognitive context. These systems align what is shown, how it is delivered, and when—based on what users can perceive and need in the moment. To achieve this, my work focuses on three interconnected thrusts:

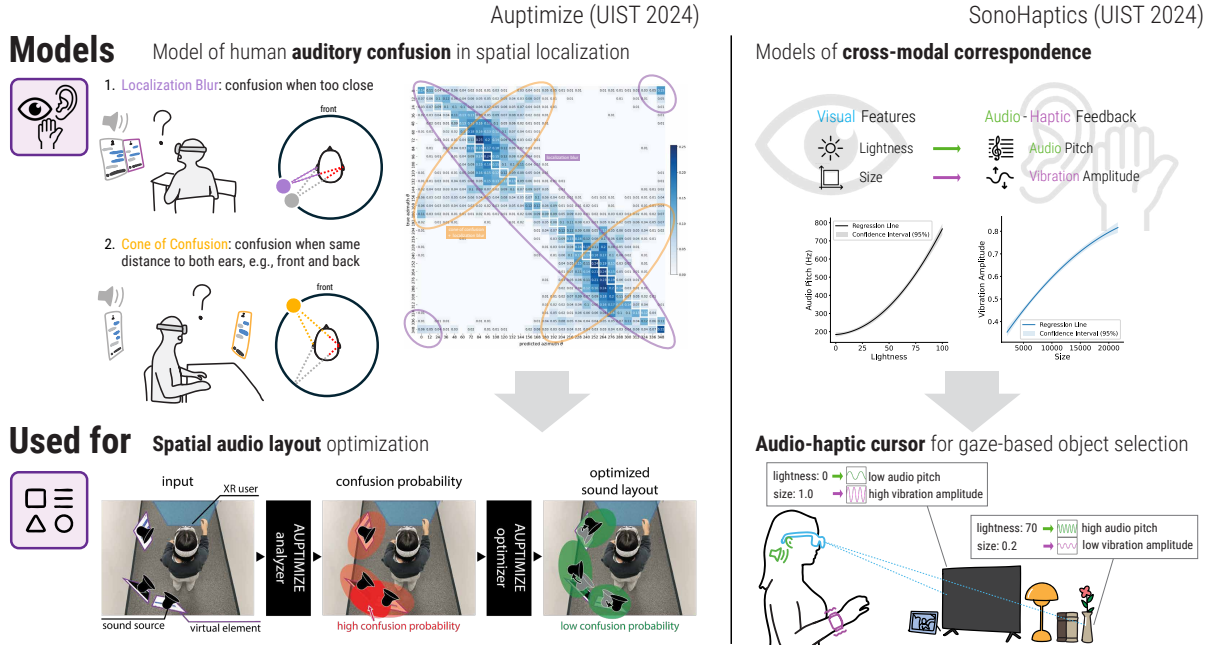


Figure 3: Models of multisensory perception and their applications in perceptually intelligent interfaces. Auptimize [7] constructs and uses a model of human auditory confusion to optimize spatial audio layouts. SonoHaptics [6] builds and uses cross-modal correspondence models to create audio-haptic cursors for target selection.

- (1) **Modeling multisensory perception and behavior:** predicting what users will perceive or miss in dynamic environments to mitigate sensory confusion and overload;
- (2) **Developing context-aware UI representations:** decomposing monolithic applications into fluid, recomposable UIs that adapt content, modality, and layout to current tasks and surroundings;
- (3) **Supporting persistent interaction continuity:** leveraging user history, intent, and preferences to maintain coherence across contexts.

For example, in a cooking scenario (Figure 2), the user misses a step in the recipe. To maintain interaction continuity, the system sends a reminder, selecting the auditory channel since the visual and manual channels are already occupied. Noticing that the front-facing auditory space is masked by sizzling sounds, it spatializes the audio reminder from the side. This contextual support leverages the modality and direction most likely to succeed, illustrating how perceptually intelligent UI can act with awareness, fluidity, and precision, without burdening the user with manual navigation through static layouts.

I present my past and ongoing projects that demonstrate how computational models of human perception can inform more effective XR interfaces that seamlessly integrate into users’ everyday activities. Lastly, I discuss the remaining challenges, opportunities, and future directions toward perceptually intelligent UIs.

2 Modeling Multisensory Perception & Behavior

Unlike screen-based interfaces, XR places digital elements anywhere in 3D space and embeds them in real-world contexts—a cooking direction floating above the stove or a colleague’s avatar

calling the user’s name from behind. The spatial and situated nature of XR creates new interaction challenges, inheriting human perceptual limitations and situational constraints from real-world activities. My approach builds computational models of multisensory perception and applies these models in novel interaction techniques to solve these interaction challenges (Figure 3).

Modeling Spatial Audio Localization Behavior (Auptimize [7]). Auditory cues play crucial roles in spatial XR interfaces, complementing the limited field of view for notifications [2], guidance, or social interactions. However, humans have two key **spatial localization errors**—*localization blur* (when sound sources are too close for accurate differentiation) and *cone of confusion* (when sounds are equidistant from both ears)—which carry over to digital soundscapes. These limitations introduce interaction friction, causing confusion when finding a notification, world-embedded information, or a person in a virtual conference.

To overcome these challenges, I developed *Auptimize*, a computational approach to improving spatial audio cue localization. I built a **computational model that predicts confusion probabilities of auditory localization** for any given layout of XR elements. The model captures both localization blur through probability distributions and cone of confusion through geometric analysis of elements positioned on the same confusion cone. I then developed an **integer programming optimization** that uses this model to automatically reposition virtual audio sources—leveraging the ventriloquist effect to decouple audio from visual locations—while preserving meaningful spatial relationships and maintaining element ordering. The results demonstrated the effectiveness of this computational approach in reducing perceptual errors by up to 25.2% relative to traditional co-located audiovisual placement.

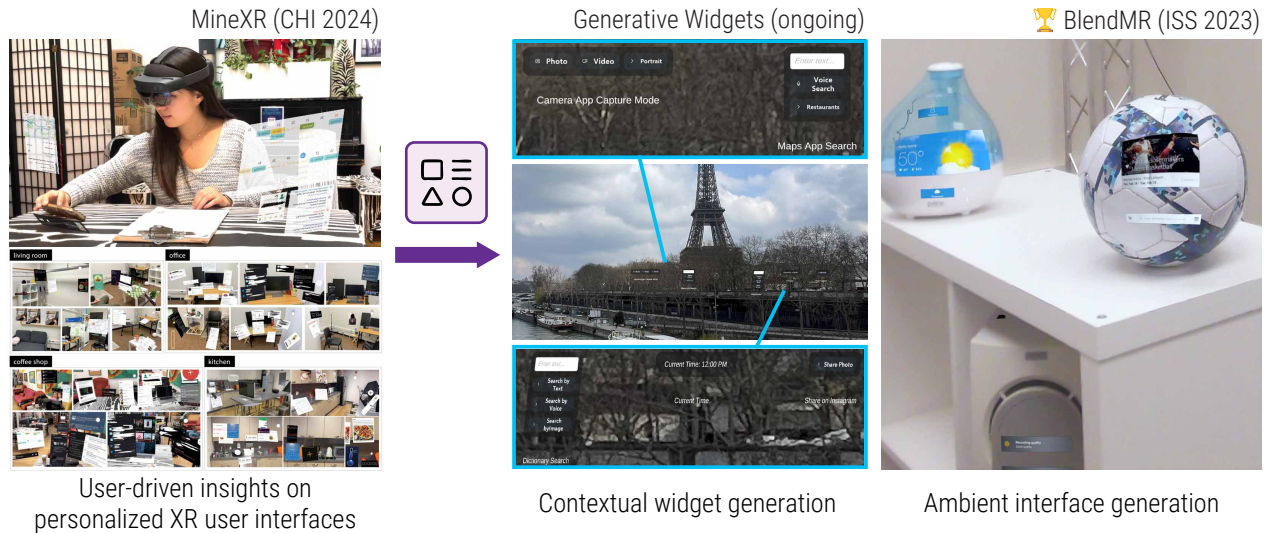


Figure 4: Approaches to developing context-aware UI representations. User-driven insights from MineXR [8] encouraged fluid, recomposable widget generation and ambient interface generation with BlendMR [9].

Cross-modal Correspondence for Channel Substitution (SonoHaptics [6]). Situational impairments in XR frequently make primary sensory channels unavailable—hands are occupied with physical tasks [10], visual displays may not exist on lightweight devices, or environmental factors compromise specific modalities. In this work, I explored scenarios where visual feedback channels are unavailable or insufficient, investigating how alternative sensory channels can provide equivalent information.

Drawing on cross-modal correspondence effects (exemplified by the “bouba/kiki” effect [12]), I contribute a quantification of how visual properties naturally map to other modalities. Using regression models, I confirmed **significant mappings between lighter visual colors to higher audio pitch**, and **between larger visual sizes and greater vibration amplitude** on the wristband. I applied these empirically-derived models to create *SonoHaptics*, an **audio-haptic cursor for gaze-based object selection**, providing feedback that assists users in selecting objects using noisy eye tracking—a common interaction in XR scenarios like looking up product information. Results showed that this short, abstract audio-haptic feedback conveyed as much information as speech descriptions, while improving selection accuracy for small, cluttered objects by 16.3% relative to no feedback. This demonstrates how computational models of cross-modal perception can create effective alternatives when primary channels are compromised.

3 Developing Context-Aware UI Representations

While perceptual models can determine what users can effectively process, XR systems still require structural frameworks to express information at the appropriate granularity and modality for each situation. My research develops systems that move beyond fixed app layouts toward **fluid, recomposable interfaces** that automatically configure based on user goals, environmental constraints, and situational factors (Figure 4).

Mining User Needs (MineXR [8]): I created a comprehensive dataset of how users envision personalized XR interfaces. The dataset revealed that (1) users want *specific task-related functionalities* with high recomposability rather than monolithic apps, and (2) users naturally organize interfaces into primary (task-focused), peripheral (occasional attention), and ambient (environmental) categories. These insights guided the development of my context-aware UI representations.

Contextual Widget Generation (ongoing): Building on MineXR’s insights, I am developing a system that automatically generates contextually appropriate XR widgets by decomposing applications into **hierarchical functionality trees organized by user goals**. The system uses vision language models to analyze users’ egocentric view and selects relevant functionality tree nodes based on task usefulness, cognitive state, environment compatibility, and activity compatibility. It then generates primary and peripheral **widgets that intelligently group functionalities across application boundaries**—for example, in Figure 4 (middle), curating camera capture modes (photo, video, portrait) into one widget and combining maps search with multiple input modalities plus restaurant shortcuts for primary tasks (top), while placing dictionary search, current time, and Instagram sharing in the periphery (bottom) for traveling. This approach promises to reduce the burden of XR UI management across diverse contexts while preventing visual clutter and situational impairment by presenting contextually relevant functionalities at **appropriate granularity and modality**.

Ambient Interface Generation (BlendMR [9]). To advance ambient UI representations of XR, I contributed to developing a system that provides situational support by transforming everyday objects into ambient information displays that provide contextual awareness without overwhelming users’ primary tasks. BlendMR uses geometry processing and integer programming optimization to map 2D application elements onto suitable physical object surfaces, optimizing for geometric suitability, viewing direction, element importance,

and hierarchical relationships. BlendMR’s context-aware, ambient UI representations reduce perceived visual clutter and distraction by 42.1% and 39.8%, compared to traditional 2D floating windows while maintaining information accessibility.

4 Supporting Persistent Interaction Continuity

Beyond understanding what users can perceive in the moment, my research develops systems that maintain awareness of user history and intent across temporal and spatial boundaries to enable seamless digital assistance (Figure 5).

Interacting with the Past (RealityReplay [4]). To enable access to digitized personal history across time, I developed *RealityReplay*, a system that continuously builds a **history of important events in users’ 360° surroundings** through semantic segmentation and saliency prediction. This enables users to replay any significant moments that they missed due to limited sight or lack of attention.

Intent-Grounded AI Assistance (Persistent Assistant [3]). Without persistent grounding of user intent, repeated interactions with wearable AI assistants can be cumbersome. Imagine a user with dietary restrictions having to repeat verbal queries in public for every product (e. g., “Is this gluten-free, dairy-free, and low in sugar?”) and listening to the assistant’s verbal response every time. This process easily becomes tedious because current AI assistant interactions are mostly turn-based verbal conversations.

I leveraged persistent grounding of user intent in creating *Persistent Assistant* that enables **intuitive and instinctive human-AI assistant interaction** (Figure 5, right). Only once, users verbally establish their intent at the beginning of a longitudinal task session, e. g., stating “I’m looking for gluten-free, dairy-free, and low-calorie products” at the supermarket entrance. From then on, throughout the session, users can simply use gaze+pinch gesture to query about an item and receive haptic encoding of the response based on the intent, in less than 1 second. Enabled by persistent interaction continuity, the haptic-centered interaction felt “intuitive” and “instinctive,” unlocking the benefits of multisensory interactions with AI assistants beyond the traditional episodic verbal ones.

5 Future Directions & Conclusion

My dissertation work establishes the foundation for perceptually intelligent user interfaces by interconnecting the three core thrusts toward adaptive support aligned with users’ perceptual capabilities, contextual constraints, and evolving intentions.

My ongoing work bridges these thrusts through systems that model the dynamic interplay between perception and interaction. I am developing a **computational rationality agent** [11] that provides adaptive guidance based on the simulation of embodied audiovisual search behavior, linking perceptual models (Thrust 1) with adaptive UI decisions (Thrust 2) and persistent goal-driven interaction (Thrust 3). Similarly, Contextual Widget Generation demonstrates how perceptual models can drive real-time interface adaptation, creating a feedback loop where the interface adapts to, and reshapes, user behavior.

This research establishes foundations for XR systems that automatically adjust their interface complexity, timing, and modality

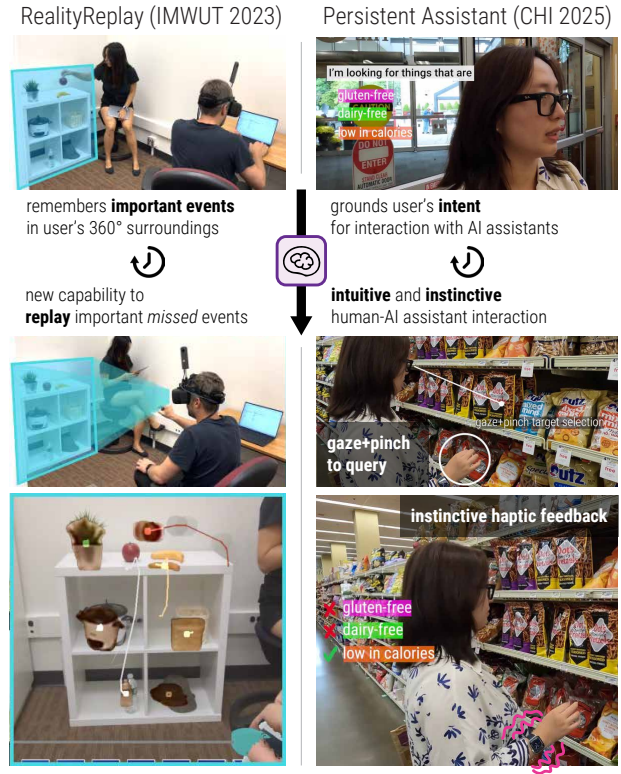


Figure 5: Systems that enable continuous interaction across time and space. RealityReplay [4] allows users to revisit past events, and Persistent Assistant [3] provides intuitive, intent-grounded interaction with AI assistants, enhancing and leveraging the continuity of the user experience

based on what users can currently handle—applicable across manufacturing training, medical procedures, educational settings, and various devices from smart glasses to immersive headsets and cars.

These foundations also transfer beyond XR, for example, to human-robot collaboration, where mutual understanding between human and machine is essential; accessible interfaces, which adapt to sensory or situational impairments; and behavioral interventions in healthcare—anywhere systems must understand and respond to human state and constraints.

Looking ahead, I plan to expand my framework to support a broader range of multisensory perception and interaction modalities, evolving alongside advances in computational capabilities, e. g., artificial intelligence. Real-world validation through longitudinal “in-the-wild” studies [1] will be critical for evaluating generalizability across diverse user populations and contexts. Additionally, enabling broad adoption requires ethical and privacy-aware design, including privacy-preserving architectures [5] and user control mechanisms for handling sensitive perceptual and behavioral data.

By creating interfaces that understand not just what users can perceive, but how their needs evolve over time and space, this research moves toward a future where technology seamlessly integrates into human experience—supporting rather than overwhelming our natural perceptual and cognitive capabilities.

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